Total No. of Questions : 4]

SEAT No. :

PD91

[Total No. of Pages : 1

[6410]-412

T.E. (Chemical Engineering) (Insem)

TRANSPORT PHENOMENA

(2019 Pattern) (Semester - II) (309350)

Time : 1 Hour] *[Max. Marks : 30*

Instructions to the candidates:

1) *Answer Q1 or Q2, Q3 or Q4.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Figures to the right indicates full marks.*
4) *Assume suitable data, if necessary.*

Q1) a) List the most used boundary conditions of Viscous flow problems. **[5]**

b) A small capillary tube with an inside diameter of 2.2×10^{-3} m and length of 0.317 m is being continuously used to measure the flowrate of the liquid having density of 990 kg/m^3 and $\mu = 1.13\times10^{-3}$ Pa.sec. The velocity of the liquid is 0.275 m/sec. Calculate the pressure drop. **[10]**

OR

Q2) Derive the expression for momentum flux, velocity, maximum velocity and average velocity for flow through annulus. **[15]**

Q3) A furnace wall is exposed to hot gases at 1100 K. The wall consists of 0.12 m of the fire brick and 0.25 m of common brick. Heat transfer coefficients on the hot side are 3000 W/m^2.K and 22 W/m^2.K on the outside. Ambient air is at 300 K. What is the heat transfer rate per square meter of wall and temperature at the interface of the two bricks? Let us consider k = 0.138 W/m.K for both materials. **[15]**

OR

Q4) a) Derive an expression for heat flux for a variable heat conductivity. **[10]**

b) List down the comparison for Forced and Free convection. **[5]**